

MAHMOUD ELLOUMI

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PROFILE

Computer Science and Applied Mathematics engineering student at ENIS. Across two internships and several projects, I have built full-stack applications powered by LLMs: conversational and data assistants running locally with Ollama, document extraction with Mistral AI, and automated workflows with n8n. Seeking a **final-year internship (PFE)** to design **AI agents** that solve real business needs.

EXPERIENCE

- **Full-Stack & AI Developer Intern – IDSOFT (ID Software Solutions)** Sfax | Jul. – Aug. 2026
AI-enhanced after-sales service (SAV) management app React, Spring Boot, PostgreSQL, Ollama
 - Web application covering the full after-sales intervention lifecycle, with three portals: customer, technician, manager.
 - Local AI with Ollama (no data sent to third parties): a conversational troubleshooting assistant that creates requests, and a natural-language data assistant (revenue, interventions, technician workload).
 - Statistics dashboard; security with JWT, TOTP two-factor authentication, BCrypt and role-based access control.
- **Mobile Developer Intern – LOGICOM Informatique** Sfax | Jul. – Aug. 2025
Product scanning mobile app Flutter
 - Flutter mobile app that scans a product’s QR code and displays its available prices and details.
 - Connected to the company database to retrieve prices and product records.

PROJECTS

- **SPARKY – Intelligent Business Document Processing (PFA)** May 2026
Team of 3, Scrum Angular, FastAPI, Mistral AI, PostgreSQL, n8n
 - Automated digitization, classification and data extraction for invoices, quotes, purchase orders and delivery notes, using Mistral AI-based OCR.
 - 3-tier architecture (Angular 20, asynchronous FastAPI REST API, PostgreSQL), supplier price comparison and analytics dashboard.
 - Workflows and notifications orchestrated with n8n; comparative study with a local n8n + Ollama (Qwen2.5) prototype.
- **Intelligent Traffic Monitoring** 2025
 - Real-time vehicle detection and classification on video streams with YOLO and OpenCV (Python).
 - Traffic analytics dashboard: vehicle counts and breakdown by vehicle type.
- **TERRANAUT – Drone-Based Detection of Sick Animals** 2024
 - Team project: detecting sick animals from thermal images captured by drones.
 - Applied computer vision and deep learning (object detection) to thermal image analysis.

TECHNICAL SKILLS

AI & LLMs	LLMs (Ollama, Mistral AI, Qwen), AI agents, Machine Learning, Deep Learning
Vision & Automation	OpenCV, YOLO, PyTorch, n8n
Languages	Python, Java, JavaScript/TypeScript, C
Web & Mobile	React, Angular, Spring Boot, FastAPI, Node.js, Flutter
Data & DevOps	PostgreSQL, MySQL, MongoDB, Kafka, Docker, Git, Linux, Scrum

EDUCATION

Engineering Degree in Computer Science and Applied Mathematics – ENIS, Sfax	2024 – Present
Preparatory Cycle in Technology – IPEIS, Sfax (top 100 in the national ranking)	2022 – 2024
Baccalaureate in Technical Sciences – 15 November 1955 High School, Sfax	2022

CERTIFICATIONS

NVIDIA Getting Started with Deep Learning • Cisco C Advanced • Scrum (SFC) • Hedera (HCF) • Microsoft MS-900

VOLUNTEERING (IEEE)

Photographer – IEEE ENIS SB (2026) • Treasurer & Webmaster – IEEE Cyber Security ENIS (2025)
Social Media Manager & Photographer – TSYP 13 • Organizer – Pitch Yourself for a Career

LANGUAGES & INTERESTS

Arabic: native French: upper-intermediate English: intermediate Spanish: elementary
Interests: photography, art, sports, innovation